Question ID: 0ad5012e

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Advanced Math	Nonlinear functions	Easy

Question

$$y = -\frac{1}{4}x^2 + 2x + 29$$

The given equation models a company's scheduled deliveries over 8 months, where y is the estimated number of scheduled deliveries x months after the end of May 2012, where $0 \leq x \leq 8$. Which statement is the best interpretation of the *y*-intercept of the graph of this equation in the *xy*-plane?

Answer

A. At the end of May 2012, the estimated number of scheduled deliveries was 0.

B. At the end of May 2012, the estimated number of scheduled deliveries was 29.

C. At the end of June 2012, the estimated number of scheduled deliveries was 0.

D. At the end of June 2012, the estimated number of scheduled deliveries was 29.

Correct Answer: B

Rationale

Choice B is correct. The *y*-intercept of a graph in the *xy*-plane is the point where $x = 0$. For the given equation, the *y*-intercept of the graph in the *xy*-plane can be found by substituting 0 for x in the equation, which yields $y = -\frac{1}{4}(0)^2 + 2(0) + 29$, or $y = 29$. Therefore, the *y*-intercept of the graph is $(0, 29)$. It's given that y is the estimated number of scheduled deliveries x months after the end of May 2012. Therefore, $x = 0$ represents 0 months after the end of May 2012, or the end of May 2012. Thus, the best interpretation of the *y*-intercept of the graph of this equation in the *xy*-plane is that at the end of May 2012, the estimated number of scheduled deliveries was 29.

Choice A is incorrect and may result from conceptual errors.

Choice C is incorrect and may result from conceptual errors.

Choice D is incorrect and may result from conceptual errors.